

PHASE 7B

Legacy PhonePe multi-instance safety

Phase 7 could not retire callbacks because 43 active legacy subscriptions and 26 pending orders remain. Phase 7B keeps those obligations serviceable while blocking new PhonePe checkout and serializing callback effects across three EC2 instances.

10 / 10	43	26	YES
Focused safety tests	Active records preserved	Pending orders preserved	Multi-instance ready

Phase 7 vs Phase 7B

New checkout	Any non-Razorpay mode could silently enter the PhonePe controller.	Explicit provider dispatch. Razorpay stays Razorpay; missing or unknown modes fail closed. PhonePe also requires PHONEPE_NEW_PAYMENTS_ENABLED=true .
Webhook deduplication	Read existing event, perform side effects, then insert - race-prone across instances.	Unique MongoDB event key is atomically claimed before any business mutation.
Callback/webhook overlap	Redirect callback and webhook could independently apply the same successful payment.	Payment outcomes share one hashed merchant-transaction identity across both endpoints.
Failure handling	Webhook processing errors were acknowledged with HTTP 200.	Owner marks the claim failed and returns 503; provider retry can reclaim it. Processed duplicates return 200/no-op.
Historical data	43 active subscriptions, 26 pending orders; retirement blocked.	All records remain intact and readable. No migration, cancellation, or deletion was performed.

Scope boundary

- Razorpay controller and business logic were not modified.
- Google Play, Apple billing, Redis, cron locks, rate limits, thumbnail storage, MongoDB pooling, ALB, DNS, EC2 configuration, and frontend code were not modified.
- Legacy GET /api/payment/callback and POST /api/webhooks/phonepe remain registered.

Atomic processing contract

Claim lifecycle

1. Authenticate and identify	Webhook signature is verified first. A stable event key is derived from provider event and transaction identity; payment callback and payment webhook converge on one key.
2. Ensure uniqueness	The path ensures the named unique index phonepe_event_key_unique exists before accepting work.
3. Atomic claim	A single findOneAndUpdate with upsert claims a new, failed, or expired-lease record. Duplicate-key contention means another instance already owns or completed it.
4. Apply effects	Only the lease owner may change order, subscription, user, mandate, or audit-event state. Payment dates are based on stable paidAt values for retry safety.
5. Complete	Owner-token guarded update marks processed. Later duplicates return HTTP 200 without provider calls or business writes.
6. Fail and retry	Exceptions mark failed and return HTTP 503 with a generic message. Nonterminal callback states also remain retryable. Logs omit provider payloads and secret-bearing errors.

Files changed

models/PhonePeEventClaim.js	Unique claim key, owner lease, processing state, attempt count, processed timestamp, safe failure code.
services/phonePeEventClaims.js	Stable identities, index assurance, atomic claim, owner-only completion, retry release.
services/phonePePolicy.js	Explicit checkout provider selection and default-off new PhonePe policy.
controllers/paymentController.js	Claim-gated legacy callback/webhook effects and retryable errors.
routes/payment.js	Explicit new-checkout dispatch; unknown modes no longer fall back to PhonePe.
services/phonePeService.js	Returns normalized provider state so pending callbacks remain retryable.
tests/phonepe-multi-instance.test.cjs	Concurrency, cross-path, retry, routing, history, index, and log-safety coverage.
.env.example / package.json	Documents default-off flag and adds focused test command.

Production configuration

- Keep **PAYMENT_GATEWAY_MODE=razorpay**.
- Set or leave **PHONEPE_NEW_PAYMENTS_ENABLED=false**. Unset also fails closed.
- Retain current **PHONEPE_*** callback verification/status credentials while legacy obligations remain.
- The MongoDB role must permit creation of the named unique claim index. The claim path ensures it before processing.

Verification and readiness

Phase 7B focused	10 / 10 PASS	Atomic contention, unique index assurance, shared callback/webhook identity, concurrent webhook and callback, failure/retry, pending state, checkout block, Razorpay routing, historical schema.
Relevant payment suites	51 / 53	Only two pre-existing annual-billing assertions fail (expected 201/409, received 500). All other PhonePe, Razorpay, simulator, mode, cancellation, and access tests pass.
Complete backend	226 / 229	Two pre-existing annual-billing failures plus one environment-only PDF worker failure because pdf-parse is unavailable to its child process. Phase 7B remains green.
Syntax and diff	PASS	Node syntax checks and git diff --check pass. No protected provider or infrastructure files changed.

Phase 7B acceptance

PASS New PhonePe initiation blocked	PASS Legacy callbacks preserved	PASS Atomic event claim
PASS Concurrent duplicate webhook	PASS Concurrent duplicate callback	PASS Payment/subscription effects once-only
PASS Retry behavior	PASS Historical records preserved	PASS Razorpay untouched

READY FOR MULTI-INSTANCE	YES
Deployment action	Not performed. Deploy through the existing process and keep new PhonePe checkout disabled.
Legacy obligations	Continue processing existing callbacks until the 43 active subscriptions and 26 pending orders are reconciled in a later phase.

Operational note

A two-minute owner lease permits recovery after a process dies. Processed duplicates are acknowledged without repeating side effects; failed or nonterminal events remain reclaimable. No PhonePe history was deleted or bulk-modified.